

Sequences and Series

Introduction

Simply put, a *sequence* is a list of numbers, a *series* is a cumulative sum (finite or infinite) of a given sequence of terms. Problems involving sequences and series are common in mathematics competition. We may want to find the general term (if it exists) of a sequence, or a sum of a sequence, some other properties such as *monotonicity* (increasing or decreasing) of a sequence, *periodicity* (or modulo some numbers) of a sequence, *divisibility*, *domain* or *range*, etc. Very often a sequence is related to *iterations* of functions, or related to some *partitions* of sets, or some *number-theoretical* properties. Thus problems involving sequences and series may also include other areas of mathematics.

We first give examples of sequences and series (how they are derived), and possibly find the general terms and sums. Then give properties of certain sequences and series.

Examples of sequences and series

Easiest examples are the arithmetical sequences and the geometric sequences. An *arithmetical* sequence $\{a_n\}$ of the form $a_1, a_2 = a_1 + d, a_3 = a_1 + 2d, \dots$, where d is the *common difference*. To sum $S = a_1 + (a_1 + d) + (a_1 + 2d) + \dots + (a_1 + (n-1)d)$ (n terms), we note (by writing the sum in reverse order) $S = (a_1 + (n-1)d) + \dots + (a_1 + 2d) + (a_1 + d) + a_1$, hence

$$2S = n(2a_1 + (n-1)d), \text{ and } S = \frac{n(2a_1 + (n-1)d)}{2}.$$

On the other hand, a *geometric sequence* is a sequence of the form $a_1, a_2 = ra_1, a_3 = r^2a_1, \dots$, where r is the *common ratio* (nonzero in general). To sum

$$S = a_1 + a_1r + a_1r^2 + \dots + a_1r^{n-1} \quad (n \text{ terms}), \text{ we note } Sr = a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^n,$$

$$\text{hence } S(r-1) = a_1(r^n - 1), \text{ or } S = a_1 \frac{(r^n - 1)}{(r - 1)} = a_1 \frac{(1 - r^n)}{(1 - r)}. \text{ In case } 0 < r < 1, \text{ say,}$$

$$\text{then } \lim_{n \rightarrow \infty} S = \frac{a_1}{1 - r}. \quad (\text{Of course in this case, the limit of the sequence is } 0.)$$

Another property we usually consider is what happens when n gets bigger and bigger, does the sequence $\{x_n\}$ have a limit, i.e., does $\lim_{n \rightarrow \infty} x_n$ exist? Typically, if a sequence x_n is increasing (decreasing) but bounded above (below), then such a limit exists. (Examples: the sequence $1, \frac{1}{2}, \frac{1}{3}, \dots$ has limit 0, yet $1 + \frac{1}{2} + \frac{1}{3} + \dots$ (*harmonic series*) does not exist (prove it). Also the sequence $\left(1 + \frac{1}{n}\right)^n$ is strictly increasing and

bounded above (by 3 say), and hence has a limit 'e'.)

We may also have a list of sequences comparison $A_n \leq B_n \leq C_n$. In case both A_n and C_n go to a common limit L , then so is B_n .

Example. n^2 ($n \geq 4$) positive numbers lined up as n rows and n columns:

$$\begin{array}{ccccccc} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn}, \end{array}$$

with each row forms an arithmetical sequence, and each column forms a geometrical sequence, and all columns are of the same common ratio. Suppose $a_{24} = 1$, $a_{42} = \frac{1}{8}$

and $a_{43} = \frac{3}{16}$. Find $a_{11} + a_{22} + \dots + a_{nn}$.

Let the common difference of the first row be d and the common ratio of each column be q , then the common ratio of the fourth row is dq^3 , also

$a_{41} = 2a_{42} - a_{43} = \frac{1}{16}$, $a_{44} = 2a_{43} - a_{42} = \frac{1}{4}$, and each column forms a geometric

sequence, hence $a_{34} = \sqrt{a_{24}a_{44}} = \frac{1}{2}$, thus $q = \frac{1}{2}$. From these, we have

$a_{11} = \frac{a_{41}}{q^3} = \frac{1}{2}$, $a_{14} = \frac{a_{44}}{q^3} = 2$. And $d = \frac{a_{14} - a_{11}}{3} = \frac{1}{2}$. From these, we have

$a_{kk} = a_{1k}q^{k-1} = (a_{11} + (k-1)d)q^{k-1} = \frac{k}{2^k}$. Thus

$$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n}.$$

Further $\frac{S}{2} = \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^3} + \dots + \frac{n}{2^{n+1}}$. Taking the difference, get

$$\frac{S}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^3} + \dots + \frac{1}{2^n} - \frac{n}{2^{n+1}} = \frac{1}{2} \cdot \frac{1 - \frac{1}{2^n}}{1 - \frac{1}{2}} - \frac{n}{2^{n+1}} = 1 - \frac{1}{2^n} - \frac{n}{2^{n+1}}. \text{ Finally}$$

$$S = 2 - \frac{1}{2^{n-1}} - \frac{n}{2^n}.$$

Example. From the sequence $1, \frac{1}{2}, \frac{1}{2^2}, \dots, \frac{1}{2^n}, \dots$, is it possible to find an infinite geometric series whose sum is $\frac{1}{5}$.

Suppose yes, let the infinite geometric series be $\{a_n\}$ with $a_1 = \frac{1}{2^k}$ and the common ratio be $q = \frac{1}{2^t}$, then the sum is $\frac{1/2^k}{(1-1/2^t)} = \frac{1}{5}$. Thus $5 = 2^k - 2^{k-t}$. When $k=0$, the equation cannot hold, when $k \neq 0$, 2^k is even, hence 2^{k-t} is odd, thus $k-t=0$, hence $2^k = 6$, impossible again.

A *sequence* may also be considered as a function f defined for every nonnegative integer n , denoted by $x_n = f(n)$. Usually it is defined as an equation of the form

$$x_n = F(x_{n-1}, x_{n-2}, \dots),$$

which is a form of *functional equation*. A functional equation of the form

$$x_n = px_{n-1} + qx_{n-2} \quad (q \neq 0)$$

is a (homogeneous) *linear difference equation of order 2* (with constant coefficients).

It is also a *recurrence relation*. To solve it, try $x_n = \lambda^n$, plug into the equation, get

$\lambda^n = p\lambda^{n-1} + q\lambda^{n-2}$. If $\lambda \neq 0$, get $\lambda^2 = p\lambda + q$, or $\lambda^2 - p\lambda - q = 0$. This is the *characteristic equation* of the difference equation. For distinct roots λ_1 and λ_2 , we

try $x_n = a\lambda_1^n + b\lambda_2^n$, with a and b determined by the initial values x_0 and x_1 . For

$\lambda_1 = \lambda_2 = \lambda$, we try $x_n = (a + bn)\lambda^n$. Higher order equations are considered likewise.

Example. The Fibonacci sequence is defined by $F_0 = 0, F_1 = 1, F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 2$. Its characteristic equation is $x^2 - x - 1 = 0$, with roots

$$\alpha = \frac{1+\sqrt{5}}{2} \quad \text{and} \quad \beta = \frac{1-\sqrt{5}}{2}. \quad \text{Assuming the general term}$$

$$F_n = A\alpha^n + B\beta^n = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n. \quad \text{Put } n=0, \text{ get } A+B=0, \text{ put } n=1,$$

$$\text{get } A-B = \frac{2}{\sqrt{5}}, \text{ hence } A = \frac{1}{\sqrt{5}} \text{ and } B = -\frac{1}{\sqrt{5}}. \text{ Thus}$$

$$F_n = A\alpha^n + B\beta^n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right).$$

It is interesting to observe all F_n are integers and yet the expression involve irrational numbers.

Taking the ratios of consecutive Fibonacci numbers, we get $\frac{1}{1}, \frac{2}{1}, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \frac{21}{13}, \dots$,

we see they actually converge to a certain number (*golden ratio*). We note

$$\frac{F_{n+1}}{F_n} = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha^n - \beta^n} = \frac{\alpha - (\beta/\alpha)^n \beta}{1 - (\beta/\alpha)^n} \rightarrow \alpha = \left(\frac{1+\sqrt{5}}{2} \right) \approx 1.618, \text{ as } n \text{ goes to infinity.}$$

Note $|\beta/\alpha| < 1$.

Example. The *Lucas* sequence is defined by $L_1 = 1, L_2 = 3, L_{n+1} = L_n + L_{n-1}, n \geq 2$.

We have the same characteristic equation as before. Hence $L_n = C\alpha^n + D\beta^n$. When

$n=1$, get $1 = C\alpha + D\beta$, when $n=2$, get $3 = C\alpha^2 + D\beta^2$, simplify to get

$C = D = 1$. Hence $L_n = \alpha^n + \beta^n$. Of course, the Fibonacci and the Lucas sequences

are closely related.

Examples. F_n and L_n are closely related, and may be defined for negative integers.

We have for instances, the following relations.

$$(i) \quad F_{-n} = -(\alpha\beta)^{-n} F_n = (-1)^{n-1} F_n, \quad L_{-n} = (-1)^n L_n.$$

$$(ii) \quad 2F_{m+n} = F_m L_n + F_n L_m.$$

$$(iii) \quad L_n^2 - 5F_n^2 = 4(-1)^n.$$

$$(iv) \quad F_n^2 - F_{n-1}F_{n+1} = (-1)^{n-1}.$$

$$(v) \quad L_n^2 - L_{n-1}L_{n+1} = 5(-1)^n.$$

$$(vi) \quad (F_n, F_{n+1}) = 1, (L_n, L_{n+1}) = 1.$$

$$(vii) \quad \text{The parity of } F_n \text{ and } L_n \text{ are the same, and } (F_n, L_n) | 2.$$

$$(viii) \quad F_{m+n} = F_m F_{n+1} + F_{m-1} F_n.$$

Either we derive them directly, or using induction, or using previous results.

Example. $(F_m, F_n) = F_{(m,n)}.$

We may suppose $m > n$. From (viii) of the last example, $F_m = F_{m-n}F_{n+1} + F_{m-n-1}F_n$,

(ix), get $(F_{m-n}, F_n) | F_m$ and $(F_{m-n}, F_n) | (F_m, F_n)$. But from (vi) and (ix), get

$$(F_m, F_n) | F_{m-n} \quad \text{and} \quad (F_m, F_n) | (F_{m-n}, F_n). \quad \text{Hence} \quad (F_m, F_n) = (F_{m-n}, F_n), \quad \text{and}$$

$(m-n, n) = (m, n)$. then we descend to get the result.

We give an example to show how some recurrence relations of sequences may be derived.

$$\textbf{Example.} \quad \begin{cases} x_k = 3x_{k-1} + 4y_{k-1} \\ y_k = 2x_{k-1} + 3y_{k-1}. \end{cases}$$

We consider elements of the form $u = x + y\sqrt{2}$, where x and y are integers. Define

its *norm* as $N(u) = (x + y\sqrt{2})(x - y\sqrt{2}) = x^2 - 2y^2$. Note that norm is *multiplicative*,

namely $N(uv) = N(u)N(v)$, which can be checked by simple verification. Now we

take $x_1 = 3$, and $y_1 = 2$, then $N(3 + 2\sqrt{2}) = (3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 1$. (Also it means

that $(3, 2)$ is an integer solution of the equation $x^2 - 2y^2 = 1$. This is a *Pell's*

Equation.) Now $N(17 + 12\sqrt{2}) = N((3 + 2\sqrt{2})^2) = N(3 + 2\sqrt{2})N(3 + 2\sqrt{2}) = 1 \times 1 = 1$,

which can indeed be verified $17^2 - 2 \times 12^2 = 1$. Suppose we define

$$(3 + 2\sqrt{2})^k = x_k + y_k \sqrt{2}.$$

Then

$$\begin{aligned} (3x_{k-1} + 4y_{k-1}) + (2x_{k-1} + 3y_{k-1})\sqrt{2} &= (x_{k-1} + y_{k-1}\sqrt{2})(3 + 2\sqrt{2}) \\ &= (3 + 2\sqrt{2})^{k-1}(3 + 2\sqrt{2}) = (3 + 2\sqrt{2})^k = x_k + y_k \sqrt{2}. \end{aligned}$$

Hence by comparing the two sides, get $x_k = 3x_{k-1} + 4y_{k-1}$ and $y_k = 2x_{k-1} + 3y_{k-1}$.

We get a list of (x_k, y_k) : $(3, 2), (17, 12), (99, 70), (577, 408), (3363, 2378), \dots$. Now

observe $x_k^2 - 2y_k^2 = 1$, hence $\frac{x_k^2}{y_k^2} - 2 = \frac{1}{y_k^2}$. And $\frac{x_k}{y_k} = \sqrt{2 + \frac{1}{y_k^2}} \leq \sqrt{2 + \frac{1}{4 \times 9^{k-1}}}$.

($y_k \geq 3y_{k-1} \geq 3^2 y_{k-2} \geq \dots \geq 3^{k-1} y_1 = 2 \times 3^{k-1}$.) This means $\frac{x_k}{y_k}$ is a good

approximation of $\sqrt{2}$. We get the numbers

1.5, 1.41666..., 1.414285..., 1.4142156..., 1.4142136..., where $\sqrt{2} = 1.4142135\dots$.

Properties of Fibonacci sequence

We are still mostly interested in the Fibonacci numbers. A lot of interesting properties may be found in *Fibonacci Numbers*, (<https://mathworld.wolfram.com>).

Example. Walking up a staircase of n steps, each time one is allowed to go up one step or two steps, no turning back.

Denote by S_n ways to walk up a staircase with n steps. Then $S_1 = 1$, $S_2 = 2$, and $S_n = S_{n-1} + S_{n-2}$, $n \geq 3$.

Example. Determine the number of 0-1 sequences of length n , then have no two consecutive 1s.

Denote this number be P_n . If such a sequence ends in 1, then the second next symbol is a 0, and the rest $n-2$ symbols are arbitrary satisfying the condition. If such a sequence ends in 0, then any allowable $n-1$ symbols are arbitrary, thus $P_n = P_{n-2} + P_{n-1}$. We have $P_0 = 1, P_1 = 2, P_2 = 3$, and so on.

Example. Write down a list of Fibonacci numbers and their squares, try to get some

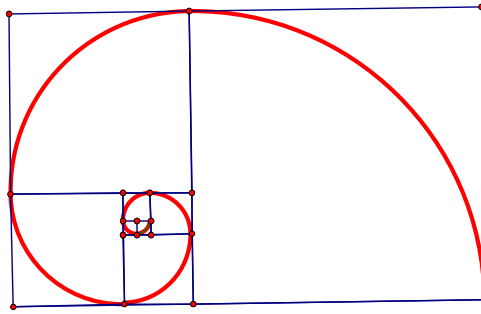
properties.

n	0	1	2	3	4	5	6	7	8	9	10	11
F_n	0	1	1	2	3	5	8	13	21	34	55	89
F_n^2	0	1	1	4	9	25	64	169	441	1156	3025	7921

Adding two consecutive terms of the squares will always get a Fibonacci number, 1, 2, 5, 13, 34, 89,.... . On the other hand, adding several consecutive terms of the squares, get the products of two Fibonacci numbers. For examples: $1+1+4=6=2\times 3$, $1+1+4+9=15=3\times 5$, $1+1+4+9+25=40=5\times 8$,

$1+1+4+9+25+64=104=8\times 13$,.... (Indeed $F_1^2 + F_2^2 + \dots + F_n^2 = F_n F_{n+1}$).

Its proof may be visualized by the picture.



Example. Let

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots + \frac{1}{1}}}} = \frac{m}{n},$$

where m and n are relatively prime integers, and the left side of the equality contains 1988 horizontal lines. Calculate $m^2 + mn - n^2$.

The fraction with k horizontal lines is denoted by $\frac{m_k}{n_k}, (m_k, n_k) = 1$. Hence

$$\frac{m_{k+1}}{n_{k+1}} = \frac{1}{1 + \frac{m_k}{n_k}} = \frac{n_k}{m_k + n_k},$$

thus $m_{k+1} = n_k, n_{k+1} = m_k + n_k$. Comparing with the Fibonacci sequence, get $m_k = F_k, n_k = F_{k+1}$. Thus

$$\begin{aligned}
& m^2 + mn - n^2 \\
&= F_{1988}^2 + F_{1988}F_{1989} - F_{1989}^2 \\
&= F_{1988}^2 + F_{1988}(F_{1988} + F_{1987}) \\
&\quad - (F_{1987}^2 + 2F_{1988}F_{1987} + F_{1988}^2) \\
&= -(F_{1987}^2 + F_{1988}F_{1987} - F_{1988}^2) \\
&= F_{1986}^2 + F_{1987}F_{1986} - F_{1987}^2 \\
&= \dots \\
&= F_2^2 + F_2F_3 - F_3^2 \\
&= 1^2 + 1 \times 2 - 2^2 = -1.
\end{aligned}$$

Example. Determine the largest possible value of $m^2 + n^2$, where m and n are integers between 1 and 1981, and $(n^2 - mn - m^2)^2 = 1$.

If $(n^2 - mn - m^2)^2 = 1$, then $n^2 - mn - m^2 = \pm 1$. Hence $n^2 = mn + m^2 \pm 1 \geq m^2$, or $n \geq m$, equality holds if and only if $m = n = 1$. When $(m, n) \neq (1, 1)$, $n > m$. In this case when (m, n) is a solution, then

$$\begin{aligned}
1 &= (n^2 - mn - m^2)^2 \\
&= ((n - m)^2 - m(n - m) - m^2)^2 \\
&= (m^2 - m(n - m) - (n - m)^2)^2.
\end{aligned}$$

Thus if $(m, n) \neq (1, 1)$ is a solution, then so is $(n - m, m)$ a solution. After finitely many steps, get $(1, 1)$. Conversely starting from $(m, n) = (1, 1)$, via $(n, m + n)$ we get all solutions. Easy to see, (m, n) are adjacent Fibonacci numbers:

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, ...

Thus the largest possibly $m^2 + n^2 = 987^2 + 1597^2 = 3524578$.

We have a very interesting property of the Fibonacci sequence, namely the *Zeckendorf Representation*: A positive integer N can be expressed *uniquely* as a sum of distinct Fibonacci numbers containing no neighbors:

(Here Fibonacci numbers are defined slightly different: $F_1 = 1, F_2 = 2, F_{n+2} = F_{n+1} + F_n, n \geq 1$. Indeed $1 = F_1 = 1, 2 = F_2 = 10$ (its Zeckendorf representation), $3 = F_3 = 100, 4 = F_3 + F_1 = 101, 5 = F_4 = 1000, 6 = F_4 + F_1 = 1001, 7 = F_4 + F_2 = 1010, 8 = F_5 = 10000, 9 = F_5 + F_1 = 10001, 10 = F_5 + F_2 = 10010, 11 = F_5 + F_3 = 10100, 12 = F_5 + F_3 + F_1 = 10101, \dots$)

Proof. The proof is followed by *induction* and the *greedy* method. First we check the first cases and is fine. Then we consider an integer N . If it is a Fibonacci number, then it is fine again. If not, there exists a j such that $F_j < N < F_{j+1}$. Consider $a = N - F_j < N$, by induction, it has a Zeckendorf representation. Also $a < F_{j+1} - F_j = F_{j-1}$, so the Zeckendorf representation of a does not contain F_{j-1} . Combined to get the Zeckendorf representation of N . (To prove the uniqueness part, first note F_j must be in the representation because $F_{j-1} + F_{j-3} + F_{j-5} + \dots$ simply does not add up to N , we then proceed by induction again).

(Note also that the $64 = 55 + 8 + 1$ is the Zeckendorf representation of 64. But $64 = 55 + 5 + 3 + 1 = 34 + 21 + 8 + 1 = 34 + 21 + 5 + 3 + 1 = 34 + 13 + 8 + 5 + 3 + 1$ as sum of Fibonacci numbers are not, since 5,3 or 34,21 are consecutive Fibonacci numbers.)

Example. The set of positive integers $N = \{1, 2, \dots, n, \dots\}$ is the union of two disjoint subsets $\{f(1), f(2), \dots, f(n), \dots\}$ and $\{g(1), g(2), \dots, g(n), \dots\}$ where $f(1) < f(2) < f(3) < \dots < f(n) < \dots$ and $g(1) < g(2) < g(3) < \dots < g(n) < \dots$ and $g(n) = f(f(n)) + 1$, for all $n \geq 1$. Find $f(240)$.

(IMO 1978) We try a few cases. Indeed $g(1) = f(f(1)) + 1 > 1$, hence $f(1) = 1$, and $g(1) = f(f(1)) + 1 = f(1) + 1 = 2$. Thus $f(2) \geq 3$, $f(3) > f(2) \geq 3$, so $f(3) \geq 4$. So $g(2) = f(f(2)) + 1 \geq f(3) + 1 \geq 4 + 1 = 5$. Fill in the gaps, we must have $g(2) = 5$, and then $f(2) = 3$, $f(3) = 4$. In the same way we have $f(4) \geq 6$ and $g(3) = f(f(3)) + 1 \geq 6 + 1 = 7$. Fill in the gap again, we have $g(3) = 7$ and $f(4) = 6$. We can go on again and try to get some formulae or hints.

Indeed there is a proof by using the Zeckendorf representation.

Indeed let $n = a_k a_{k-1} \dots a_0$ be its Zeckendorf representation. Adjoin a 0 to get $f(n) = a_k a_{k-1} \dots a_0 0$. (First case) then $f(n) = a_k a_{k-1} \dots a_0 0$ if it ends with even number of 0s. (Second case) If it ends with an odd number of 0s, say of the form $\dots \underbrace{100\dots 0}_{\text{odd number of 0s}}$ then we subtract 1, corresponds to $\dots \underbrace{010101\dots 01}_{\text{subtract 1}}$. And this is the $f(n)$.

This procedure gives all numbers ending in an even number of 0s in an increasing

order, hence it computes $f(n)$. On the other hand, if $f(n) = a_k a_{k-1} \dots a_0$, ends with even number of 0s, then $f(f(n)) + 1 = a_k a_{k-1} \dots a_0 0$, which ends in odd number of 0s,

(all possible cases) i.e., it computes $g(n)$. For example, if $n = F_{i_k} F_{i_{k-1}} \dots F_{i_1}$, then

$$f(n) = F_{i_k+1} + F_{i_{k-1}+1} + \dots + F_{i_1+1} \quad (\text{First case}); \quad \text{or}$$

$$f(n) = F_{i_k} + F_{i_{k-1}} + \dots + F_{i_1} + \dots + F_5 + F_3 + F_1 \quad (\text{second case}). \text{ Finally}$$

$$240 = 233 + 5 + 2 = F_{12} + F_5 + F_2, \text{ hence } f(240) = F_{13} + F_5 + F_3 = 388.$$

Indeed there is another proof using the *Beatty's Theorem*. If α, β are irrational numbers satisfying $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, then the sequences $\{[n\alpha]\}$ and $\{[n\beta]\}$ are complementary (no intersection and together fill up all positive integers).

Take $\alpha = \frac{1+\sqrt{5}}{2}$, and $\beta = 1 + \alpha$, then one sees easily $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, also define

$$f(n) = [n\alpha] \quad \text{and} \quad g(n) = [n\beta] \quad \text{one shows the condition } f(f(n)) + 1 = g(n) \text{ is}$$

$$\text{satisfied. Hence } f(240) = [240\alpha] = [120(1+\sqrt{5})] = 120 + [120\sqrt{5}]. \quad \text{Since}$$

$$\sqrt{5} \approx 2.236, \quad \text{get} \quad 120\sqrt{5} \approx 264 + 4.32, \quad \text{hence} \quad [120\sqrt{5}] = 268,$$

$$f(240) = 120 + 268 = 388.$$

Example. Does there exists a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 2, f(f(n)) = f(n) + n$, and $f(n) < f(n+1)$ for all n ?

(IMO1993) Again set the Zeckendorf representation $n = F_{n_1} + F_{n_2} + \dots + F_{n_k}$, where

$$n_1 > n_2 > \dots > n_k > 0 \quad \text{and} \quad n_i - n_{i+1} > 1. \quad \text{Set} \quad f(1) = f(F_1) = F_2 = 2 \quad \text{and} \quad \text{for}$$

$$n = F_{n_1} + F_{n_2} + \dots + F_{n_k}, \quad \text{set} \quad f(n) = F_{n_1+1} + F_{n_2+1} + \dots + F_{n_k+1}. \quad \text{Clearly} \quad f(n+1) > f(n).$$

Also the condition $f(f(n)) = f(n) + n$ follows from

$$f(f(n)) = F_{n_1+2} + F_{n_2+2} + \dots + F_{n_k+2}.$$

Miscellaneous examples

Example (Oct 2018 Test). Determine all sequences $p_1, p_2, p_3, \dots$ of prime numbers for which there exists an integer k such that the recurrence relation

$$p_{n+2} = p_{n+1} + p_n + k$$

holds for all positive integers n .

The sequence must be a constant sequence. Using $p_{n+3} = p_{n+2} + p_{n+1} + k$ and $p_{n+2} = p_{n+1} + p_n + k$, by subtracting, get $p_{n+3} - 2p_{n+2} + p_n = 0$, we get the characteristic equation

$$\lambda^3 - 2\lambda^2 + 1 = 0, \text{ with roots } 1, \frac{1 \pm \sqrt{5}}{2}, \text{ hence}$$

by the method of characteristic equation, the general solution to the recurrence relation is

$$p_n = A \left(\frac{1 + \sqrt{5}}{2} \right)^n + B \left(\frac{1 - \sqrt{5}}{2} \right)^n + C,$$

where A, B, C are some constants. Furthermore, as the sequence must be periodic modulo p_1 (for there are at most p^2 possibilities for (p_{n+1}, p_n) which determines p_{n+2}), infinitely many terms of the sequence are congruent to p_1 modulo p_1 (hence equal to p_1). This forces $A = 0$ (for otherwise the sequence will eventually be strictly increasing if $A > 0$ or strictly decreasing if $A < 0$) and hence $B = 0$ (for otherwise p_n cannot be an integer). This gives us the desired conclusion.

Example (Chinese Mathematical Contest 1990). Let $E = \{1, 2, \dots, 200\}$ and $G = \{a_1, a_2, \dots, a_{100}\} \subseteq E$ with the following properties:

(i) For all $1 \leq i < j \leq 100$, $a_i + a_j \neq 201$.

(ii) $\sum_{i=1}^{100} a_i = 10080$.

Show that the number of odd members in G is a multiple of 4, also the sum of squares of members in G is a constant.

(i) Clearly the number of odd members in G cannot be $4k+1$ or $4k+3$. Assume

the number of odd members in G be $4k+2$, say $a_1, a_2, \dots, a_{4k+2}$, then $a_{4k+3}, a_{4k+4}, \dots, a_{100}$ are even numbers. Hence those even numbers not belonging to G are $201-a_1, 201-a_2, \dots, 201-a_{4k+2}$. Now the sum of all even numbers from 1 to 200 is 10100. Thus $\sum_{j=4k+3}^{100} a_j + \sum_{j=1}^{4k+2} (201-a_j) = 10100$. Also $\sum_{j=4k+3}^{100} a_j + \sum_{j=1}^{4k+2} a_j = 10080$, solve to get $\sum_{j=1}^{4k+2} a_j = 191 + 402k$, contradiction.

(ii) Let $a_1, a_2, \dots, a_{100}$ be members of G , then $201-a_1, 201-a_2, \dots, 201-a_{100}$ are members not in G . Hence

$$\sum_{k=1}^{200} k^2 = \sum_{i=1}^{100} a_i^2 + \sum_{i=1}^{100} (201-a_i)^2 = 2 \sum_{i=1}^{100} a_i^2 - 402 \sum_{i=1}^{100} a_i + 201^2 \times 100, \text{ hence the value of } \sum_{i=1}^{100} a_i^2 \text{ is fixed.}$$

Example (Chinese Mathematical Olympiad 1988). Let $\{a_n\}$ be a sequence, with $a_1 = 1, a_2 = 2$, and when $n \geq 1$,

$$a_{n+2} = \begin{cases} 5a_{n+1} - 3a_n & \text{if } a_n \cdot a_{n+1} \text{ even,} \\ a_{n+1} - a_n & \text{if } a_n \cdot a_{n+1} \text{ odd.} \end{cases}$$

Show that $a_n \neq 0$.

We may prove by induction $a_{2n-1} \equiv 1 \pmod{3}, a_{2n} \equiv 2 \pmod{3}$.

Example. Let $\{a_n\}$ be a sequence, with $a_1 = -1$, and when $n \geq 1$, and

$$a_{n+1} = 2a_n + \sqrt{3a_n^2 + 1}, \quad n \geq 1.$$

Show that all a_n 's are integers.

Try a few values and guess $a_{n+2} = 4a_{n+1} - a_n$ and prove it by induction. Indeed we have $a_n^2 - 4a_n a_{n+1} + (a_{n+1}^2 - 1) = 0$ and $a_{n+2}^2 - 4a_{n+2} a_{n+1} + (a_{n+1}^2 - 1) = 0$, thus a_n, a_{n+2} are (distinct) roots of the equation $x^2 - 4a_{n+1}x + (a_{n+1}^2 - 1) = 0$. From Vieta, get $a_{n+2} + a_n = 4a_{n+1}$ for all n .

Example. Let $\{a_n\}, \{b_n\}$ be sequences with the following properties:

$$\begin{aligned}a_0 &= 1, a_1 = 1, a_{n+1} = a_n + 2a_{n-1}, \quad n \geq 1, \\b_0 &= 1, b_2 = 7, b_{n+1} = 2b_n + 3b_{n-1}, \quad n \geq 1.\end{aligned}$$

Show that the two sequences have no common term except the “1”.

Try modulo 8, get 1, 1, 3, 5, 3, 5, 3, 5, ... for a_n and 1, 7, 1, 7, 1, 7, ... for b_n . Hence conjecture $a_{2n+1} \equiv 3 \pmod{8}$, $a_{2n+2} \equiv 5 \pmod{8}$ and $b_{2n+1} \equiv 1 \pmod{8}$, $a_{2n+2} \equiv 7 \pmod{8}$.

Example (Putnam 1963). Let $f(n)$ be a strictly increasing sequence with $f(2) = 2$ and when m, n relatively prime, $f(mn) = f(m)f(n)$. Show that $f(n) = n$ for all n .

First $f(1) = 1$, be taking $m = n = 1$. Now $f(3) > 2$, take $m = 3, n = 7$, get $f(3)f(7) = f(21) < f(22) = f(2)f(11) = 2f(11) < 2f(14) = 2f(2)f(7) = 4f(7)$, get $f(3) < 4$, thus $f(3) = 3$. Assume now n_0 is the smallest number such that $f(n_0) > n_0$, that means $n_0 \geq 4$, with $f(1) = 1, f(2) = 2, \dots, f(n_0 - 1) = n_0 - 1$, but $f(n_0) > n_0$. This will imply $f(n) > n$ for $n \geq n_0$. Now if n_0 is odd, then 2 and $n_0 - 2$ relatively prime, we have $f(2(n_0 - 2)) = f(2)f(n_0 - 2) = 2(n_0 - 2)$, but we should have $f(2(n_0 - 2)) > 2(n_0 - 2)$. If n_0 is even, consider 2 and $n_0 - 1$, get $f(2(n_0 - 1)) = f(2)f(n_0 - 1) = 2(n_0 - 1)$, again the same conclusion!

Example. Show that the term $a_n = \sum_{k=0}^n \binom{2n+1}{2k+1} 2^{3k}$, $n \geq 0$, is never divisible by 5.

Check that $a_0 = 1, a_1 = 11$, and

$$\begin{aligned}a_n &= \frac{(2\sqrt{2}+1)^{2n+1} + (2\sqrt{2}-1)^{2n+1}}{4\sqrt{2}} \\&= \frac{(2\sqrt{2}+1)(9+4\sqrt{2})^n + (2\sqrt{2}-1)(9-4\sqrt{2})^n}{4\sqrt{2}},\end{aligned}$$

with $(9+4\sqrt{2}) + (9-4\sqrt{2}) = 18$ and $(9+4\sqrt{2})(9-4\sqrt{2}) = 49$.

We thus have a recursive relation

$$a_{n+1} = 18a_n - 49a_{n-1} \equiv 3a_n + a_{n-1} \pmod{5}.$$

Checking the remainder mod 5, we have 1, 1, 4, 3, 3, 2, 4, 4, 1, 2, 2, 3, 1, 1, we thus have a periodic sequence $a_{n+12} \equiv a_n \pmod{5}$.

(Alternative Solution) Taking modulo 5, we have $S_n = \sum_{k=0}^n \binom{2n+1}{2k+1} (-2)^k$ not divisible

by 5. Expanding $(1+i\sqrt{2})^{2n+1} = R_n + i\sqrt{2}S_n$, (even and odd terms), where

$$R_n = \sum_{k=0}^n \binom{2n+1}{2k} (-2)^k. \text{ Multiplying by their respective conjugates, get}$$

$$3^{2n+1} = R_n^2 + 2S_n^2. \text{ Taking modulo 5, get } 3(-1)^n \equiv R_n^2 + 2S_n^2 \pmod{5}. \text{ But if}$$

$$S_n \equiv 0 \pmod{5}, \text{ then } R_n^2 \equiv \pm 3 \pmod{5}, \text{ impossible.}$$

Example. Let k be a positive integer. Find the number of non-negative integers n less than or equal to 10^k satisfying the following conditions:

- (i) n is divisible by 3;
- (ii) Each decimal digit of n is one of the digits 2, 0, 1 or 7.

As 10^k is not divisible by 3, we have to count the numbers with no more than k digits, from 0 to $\underbrace{99\dots9}_{k \text{ digits}}$. (Note 0 is included). Indeed we count the numbers of form

$\overline{a_1 a_2 \dots a_k}$, with $a_i = 2, 0, 1$ or 7 . But 3 divides $\overline{a_1 a_2 \dots a_k}$ if and only if 3 divides $a_1 + a_2 + \dots + a_k$, hence we have to count those tuples $(a_1, a_2, \dots, a_k)$ such that 3 divides $a_1 + a_2 + \dots + a_k$.

Define $P(x) = (x^2 + 1 + x + x^7)^k$. Then

$$P(x) = (x^2 + 1 + x + x^7)^k = \sum x^{a_1 + a_2 + \dots + a_k}.$$

The number of tuples $(a_1, a_2, \dots, a_k)$ such that 3 divides $a_1 + a_2 + \dots + a_k$ is the sum of coefficients of those terms of $P(x)$ whose corresponding degrees are divisible by 3. Let the required number of T . Now let ξ be a complex cube root of 1, we have then $\xi^3 = 1$ and $\xi^2 + \xi + 1 = 0$. Note also that if q is a multiple of 3, then $\xi^{2q} + \xi^q + 1 = 3$. But if q is not a multiple of 3, then $\xi^{2q} + \xi^q + 1 = 0$. Using this

observation, we have $T = \frac{P(1) + P(\xi) + P(\xi^2)}{3}$. (*Residue Siever*). But from the

definition of $P(x)$, we have

$$T = \frac{P(1) + P(\xi) + P(\xi^2)}{3} = \frac{4^k + \xi^k + \xi^{2k}}{3} = \begin{cases} \frac{4^k + 2}{3}, & \text{if } 3|k \\ \frac{4^k - 1}{3}, & \text{otherwise.} \end{cases}.$$

(Alternative Solution by Lee Ker Yang, sketched) Denote by a_k, b_k, c_k the numbers of integers with k or fewer digits, with digit sums equal 0, 1 or 2 (mod 3) respectively, and those integers contain only the digits 2, 0, 1 or 7 only. Note $1 \equiv 7 \pmod{3}$.

To count a_{k+1} , we add 0 to a sequence with digit sum 0 (mod 3), or 2 to a sequence with digit sum 1 (mod 3), or 1, 7 to a sequence with digit sum 2 (mod 3). Thus

$$a_{k+1} = a_k + b_k + 2c_k.$$

Likewise

$$b_{k+1} = 2a_k + b_k + c_k,$$

$$c_{k+1} = a_k + 2b_k + c_k.$$

We also have

$$\begin{aligned} a_1 &= 1, b_1 = 2, c_1 = 1, \\ a_2 &= 5, b_2 = 5, c_2 = 6, \\ a_3 &= 22, b_3 = 21, c_3 = 21. \end{aligned}$$

Note if we let $d_k = a_k + b_k + c_k$, then $d_k = 4^k$ (all possible integers with digits 2, 0, 1 or 7).

We may prove by induction that for $k = 3m+1$, $a_k = c_k = \left\lfloor \frac{4^k}{3} \right\rfloor, b_k = \left\lceil \frac{4^k}{3} \right\rceil$; for

$$k = 3m+2, \quad a_k = b_k = \left\lfloor \frac{4^k}{3} \right\rfloor, c_k = \left\lceil \frac{4^k}{3} \right\rceil; \quad \text{for } k = 3m, \quad a_k = \left\lceil \frac{4^k}{3} \right\rceil, b_k = c_k = \left\lfloor \frac{4^k}{3} \right\rfloor.$$

Hence $a_k = \left\lfloor \frac{4^k}{3} \right\rfloor = \frac{4^k - 1}{3}$, when $3 \nmid k$ and $a_k = \left\lceil \frac{4^k}{3} \right\rceil = \frac{4^k + 2}{3}$, when $3|k$. One

may verify these statements are consistent with the original proof.

Exercises

1. Grouping the natural numbers as follows:

$$(1), (2, 3), (4, 5, 6), (7, 8, 9, 10), (11, 12, 13, 14, 15), \dots,$$

find the sum of the n^{th} group.

1. The number of elements in the first k^{th} group is $1 + 2 + \dots + k = \frac{k(k+1)}{2}$. Thus in

the n^{th} group (there are n elements) the first element is $\frac{(n-1)n}{2} + 1$, and the last

element is $\frac{(n-1)n}{2} + 1 + n - 1 = \frac{n^2 - n + 2n}{2} = \frac{n(n+1)}{2}$. And the sum is

$$\frac{n}{2} \left(\frac{(n-1)n}{2} + 1 + \frac{n(n+1)}{2} \right) = \frac{n}{2} \left(\frac{2n^2 + 2}{2} \right) = \frac{n(n^2 + 1)}{2}.$$

2. A power of 2 cannot be expressed as the sum of k ($k \geq 2$) consecutive positive integers. Otherwise it can be done.

2. Let $n = m + (m+1) + \dots + (m+k)$, where m and k are positive integers, hence

$$n = \frac{(k+1)(2m+k)}{2}. \text{ If } k \text{ is odd, then } (2m+k) \mid n; \text{ if } k \text{ is even, then } (k+1) \mid n; \text{ in}$$

both cases n contains odd divisors. Hence not a power of 2.

On the other hand, if $n = 2^a(2b+1)$, $b \geq 1$. If $b < 2^a$, then n is the sum of $2b+1$ numbers:

$$n = (2^a - b) + (2^a - b + 1) + \dots + (2^a + b).$$

If $b \geq 2^a$, then n is the sum of 2^{a+1} numbers:

$$n = (b - 2^a + 1) + (b - 2^a + 2) + \dots + (b + 2^a).$$

3. (Putnam 1979) Real numbers satisfying

$$x_n = \frac{x_{n-2}x_{n-1}}{2x_{n-2} - x_{n-1}}, n \geq 3.$$

Find a necessary and sufficient conditions on x_1 and x_2 so that there exist infinitely many n with x_n integers.

3. $\frac{1}{x_n} = \frac{2}{x_{n-1}} - \frac{1}{x_{n-2}}$. Put $r_n = \frac{1}{x_n}$, get $r_n - r_{n-1} = r_{n-1} - r_{n-2}$ arithmetical progression

with first term $r_1 = \frac{1}{x_1}$, common difference $d = \frac{1}{x_2} - \frac{1}{x_1} = \frac{x_1 - x_2}{x_1 x_2}$. Hence

$r_n = r_1 + (n-1)d$, get $x_n = \frac{1}{r_n} = \frac{x_1 x_2}{x_2 + (n-1)(x_2 - x_1)}$. If $x_1 \neq x_2$, then $|x_n| < 1$

when n large and the condition cannot be fulfilled. Hence $x_1 = x_2$, and must be an integer.

4. Let $a_1 = 2, a_{k+1} = 3a_k + 1, k \geq 1$. Find the sum of $a_1 + a_2 + \dots + a_n$.

4. Note $a_{k+1} + \frac{1}{2} = 3(a_k + \frac{1}{2})$, a geometric progression. Simplify to get

$$a_1 + a_2 + \dots + a_n = \frac{5}{4}(3^n - 1) - \frac{n}{2}.$$

5. Show that $\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \dots + \sin \frac{(n-1)\pi}{n} = \cot \frac{\pi}{2n}$.

5. Let $z = e^{i\pi/2n}$, then $\sin(k\pi/n) = \text{Im } z^{2k}$ and $z^{2n} = -1$. We get

$$\begin{aligned} \sum_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right) &= \text{Im} \sum_{k=1}^{n-1} z^{2k} \\ &= \text{Im} \frac{z^2 - z^{2n}}{1 - z^2} \\ &= \text{Im} \frac{z^2 + 1}{1 - z^2} \\ &= \text{Im} \frac{z + z^{-1}}{z^{-1} - z} \\ &= \text{Im} \frac{2 \cos(\pi/2n)}{-2 \sin(\pi/2n)} \\ &= \cot(\pi/2n). \end{aligned}$$

6. The four binomial coefficients $\binom{n}{k}, \binom{n}{k+1}, \binom{n}{k+2}, \binom{n}{k+3}$ cannot form an arithmetical progression when $k+3 \leq n$.

6. Consider two rows down and get a list

$$\begin{array}{cccc} a & a+d & a+2d & a+3d \\ 2a+d & 2a+3d & 2a+5d & \\ 4a+4d & 4a+8d & & \end{array}$$

Dividing to get two '4' and shows impossibility.

7. Evaluate $\frac{1}{1 \cdot 2} + \frac{2}{1 \cdot 3} + \dots + \frac{F_{n+1}}{F_n \cdot F_{n+2}}$, where F_k is the k^{th} Fibonacci number.

7.

$$\begin{aligned}\sum_{k=1}^n \frac{F_{k+1}}{F_k F_{k+2}} &= \sum_{k=1}^n \left(\frac{1}{F_k} - \frac{1}{F_{k+2}} \right) \\ &= 1 + 1 - \frac{1}{F_{n+1}} - \frac{1}{F_{n+2}} \\ &= 2 - \frac{F_{n+3}}{F_{n+1} F_{n+2}}.\end{aligned}$$

8. Evaluate $\sum_{k=0}^n (F_{2^k})^{-1}$.

$$8. \sum_{k=0}^n \frac{1}{F_{2^k}} = \frac{1}{F_1} + \frac{1}{F_2} + \sum_{k=2}^n \left(\frac{F_{2^{k-1}-1}}{F_{2^{k-1}}} - \frac{F_{2^k-1}}{F_{2^k}} \right) = 3 - \frac{F_{2^n-1}}{F_{2^n}}.$$

The proof is roughly as follows. .

By induction $F_{k-1}F_m - F_k F_{m-1} = (-1)^k F_{m-k}$, $k = 1, 2, \dots, m$. Take $m = 2k$, get

$F_{k-1}F_{2k} - F_k F_{2k-1} = (-1)^k F_k$, hence $\frac{(-1)^k}{F_{2k}} = \frac{F_{k-1}}{F_k} - \frac{F_{2k-1}}{F_{2k}}$. We have a telescopic series.

9. The *arithmetic-geometric mean of Gauss*. Let $0 < a < b$. Define two sequences a_n, b_n as follows.

$$a_0 = a, b_0 = b, \quad a_{n+1} = \sqrt{a_n b_n}, b_{n+1} = \frac{a_n + b_n}{2}.$$

(i) Show that a_n is strictly increasing, b_n strictly decreasing, and $a_n < b_n$ for all n .

(ii) Show that $b_{n+1} - a_{n+1} = (b_n - a_n)^2 / 8b_{n+2}$.

(iii) Moreover, a_n and b_n tend to a same limit.

9. Routine checking.

10. Let $\{a_n\}$ be the sequence of rational numbers defined by $a_0 = 2021$, and

$a_{n+1} = a_n + \frac{2}{a_n}$, for all $n \geq 0$. Can any $a_n, n \geq 1$, be a square of a rational number?

10. We look at the sequence modulo 5. This is possible since $a_n \not\equiv 0 \pmod{5}$. Now

$$\begin{aligned}
a_0 &\equiv 1 \pmod{5}, \\
a_1 &\equiv 1+2 \equiv 3 \pmod{5}, \\
a_2 &\equiv 3+2 \cdot 3^{-1} \equiv 3+2 \cdot 2 \equiv 2 \pmod{5}, \\
a_3 &\equiv 3 \pmod{5}, \\
a_4 &\equiv 2 \pmod{5}, \\
&\dots \\
&\dots \\
&\dots
\end{aligned}$$

Thus the sequence alternates between 2 and 3 modulo 5, which are not squares modulo 5. Hence there is no square of rational numbers in this sequence.

11. Let F_n be the n^{th} Fibonacci number, that is $F_0 = 0, F_1 = 1, F_2 = 1$, and

$F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. Prove that

$$\prod_{n=2}^{\infty} \frac{F_{2n} + 1}{F_{2n} - 1} = 3.$$

11. Let L_n be the n^{th} Lucas number, that is $L_0 = 2, L_1 = 1, L_2 = 3$, and

$L_n = L_{n-1} + L_{n-2}$ for $n \geq 3$. Using induction, one may get a lot of results:

$$F_{2n} = F_n L_n, \quad F_{2n+1} - F_n L_{n+1} = (-1)^n, \quad F_{2n+1} - F_{n+1} L_n = (-1)^{n+1},$$

$$F_{2n} - F_{n-1} L_{n+1} = (-1)^{n+1}, \quad F_{2n} - F_{n+1} L_{n-1} = (-1)^n.$$

Applying the last two identities to the factors in the product, we find $\prod_{n=2}^N \frac{F_{2n} + 1}{F_{2n} - 1}$

telescopes to $\frac{F_1 L_2}{F_2 L_1} \frac{F_N L_{N+1}}{F_{N+1} L_N}$ for N even, and to $\frac{F_1 L_2}{F_2 L_1} \frac{F_{N+1} L_N}{F_N L_{N+1}}$ for N odd. Because

$$\lim_{N \rightarrow \infty} \frac{F_{N+1}}{F_N} = \lim_{N \rightarrow \infty} \frac{L_{N+1}}{L_N} = \frac{1 + \sqrt{5}}{2}, \text{ the result follows.}$$

12. Let $\{x_n\}, \{y_n\}$ be two positive real sequences defined as $x_1 = 1, y_1 = \sqrt{3}$ and

$$\begin{cases} x_{n+1} y_{n+1} - x_n = 0, \\ x_{n+1}^2 + y_n = 2, n \geq 1. \end{cases}$$

Show that the sequences are convergent and find their limits.

12. (Vietnam MO 2014). Show by induction that for all positive integers,

$$x_n = 2 \sin \frac{\pi}{3 \cdot 2^n}, y_n = 2 \cos \frac{\pi}{3 \cdot 2^n}. \text{ The limits are 0 and 2 respectively.}$$

13. Let $a > 0$ and the sequence $\{x_n\}$ is defined by $x_1 = a$, and $x_{n+1} = x_n + \frac{\sqrt{x_n}}{n^2}$ for all $n \geq 1$. Show that the sequence has a finite limit.

13. (Vietnam 2013). We have $x_{n+1} < x_n + \frac{1}{4n^4} = x_n + 2 \frac{\sqrt{x_n}}{2n^2} + \frac{1}{4n^4} = \left(\sqrt{x_n} + \frac{1}{2n^2} \right)^2$.

Therefore $\sqrt{x_{n+1}} < \sqrt{x_n} + \frac{1}{2n^2}$ for all n . Iteratively, get

$$\begin{aligned} \sqrt{x_{n+1}} &< \sqrt{a} + \sum_{i=1}^n \frac{1}{2i^2} < \sqrt{a} + \frac{1}{2} + \sum_{i=2}^n \frac{1}{2i(i-1)} \\ &< \sqrt{a} + \frac{1}{2} + \frac{1}{2} \sum_{i=2}^n \left(\frac{1}{i-1} - \frac{1}{i} \right) = \sqrt{a} + \frac{1}{2} + \frac{1}{2} \left(1 - \frac{1}{n} \right) < \sqrt{a} + 1, \end{aligned}$$

So x_n is strictly increasing and bounded above.

14. Find a bounded infinite sequence $\{x_n\}$ such that $|x_i - x_j| |i - j|^a \geq 1$ for every i, j . Here $a > 1$.

14. (IMO1991, in fact an odd USSR problem). The sequence $4\{n\sqrt{2}\}$ works. Here

$\{x\} = x - [x]$ is the fractional part of x . If p and q are integers satisfying

$$p < (4 - \sqrt{2})q, \text{ then } \left| \sqrt{2} - \frac{p}{q} \right| = \frac{|2q^2 - p^2|}{q(q\sqrt{2} + p)} > \frac{1}{4q^2}. \text{ Hence when } M > K \geq 1,$$

$$\left| \{M\sqrt{2}\} - \{K\sqrt{2}\} \right| = \left| (M - K)\sqrt{2} - L \right| > \frac{1}{4(M - K)}. \text{ Here}$$

$$L = [M\sqrt{2}] - [K\sqrt{2}] < M\sqrt{2} - K\sqrt{2} + 1 \leq (M - K)(\sqrt{2} + 1) \leq (4 - \sqrt{2})(M - K).$$

The problem is also related to the Liouville's theorem in number theory: let x be an

algebraic number of degree n , then there exists a constant c such that $\left|x - \frac{p}{q}\right| \geq \frac{c}{q^n}$ for every p and q .

15. Show that for any three infinite sequences of natural number:

$$a_1, a_2, \dots, a_n, \dots,$$

$$b_1, b_2, \dots, b_n, \dots,$$

$$c_1, c_2, \dots, c_n, \dots$$

one can find p and q such that $a_p \geq a_q, b_p \geq b_q, c_p \geq c_q$.

15. (All Russia 1961). First note there exists $i_1, i_2, \dots$ such that $a_{i_1} \leq a_{i_2} \leq \dots$, where

a_{i_1} is the smallest number among the a_i 's, and a_{i_2} is the next smallest one, and

so on. From this list, we analogously find $j_1, j_2, \dots$ such that $b_{j_1} \leq b_{j_2} \leq \dots$. Now

again from these j_s 's, we find the list $c_{k_1} \leq c_{k_2} \leq \dots$, and the conclusion follows.

16. Given positive integers $a_0, a_1, a_2, \dots$. It is given that

$a_1 > a_0, a_2 = 3a_1 - 2a_0, a_3 = 3a_2 - 2a_1, \dots$ and so on. Show that $a_{100} > 2^{99}$.

16. (All Russia 1962).

$a_2 - a_1 = 2(a_1 - a_0), a_3 - a_2 = 2(a_2 - a_1), \dots, a_{100} - a_{99} = 2(a_{99} - a_{98})$. Multiplying these

99 equations and simplify to get $a_{100} = a_{99} + 2^{99}(a_1 - a_0) > 2^{99}$. (More accurate

estimate, $a_k \geq 2^k, a_{k+1} - a_k \geq 2^k$ by induction to get $a_{100} \geq 2^{100}$).

17. Let $a_0, a_1, \dots, a_n$ be numbers, $a_0 = a_n = 0$, $a_{i-1} - 2a_i + a_{i+1} \geq 0$ for all i . Show that all terms are negative.

17. As $a_{i+1} - a_i \geq a_i - a_{i-1}$ (subsequent slope larger than earlier slope, hence (i, a_i) are all below the x -axis. Indeed for some $m \geq 1, a_{m-1} \leq 0, a_m > 0$, then $a_n - a_{n-1} \geq a_{n-1} - a_{n-2} \geq \dots \geq a_{m+1} - a_m \geq a_m - a_{m-1} > 0$, contradicting $a_n = 0$.

18. Let $a_1 = 1, a_n = a_{n-1} + \frac{1}{a_{n-1}}, n \geq 1$. Show that $a_{100} \geq 14$.

18. When $k \geq 1$, $a_k^2 = a_{k-1}^2 + 2 + \frac{1}{a_{k-1}^2}$, and as $a_k > 1$, get $a_{k-1}^2 + 2 < a_k^2 < a_{k-1}^2 + 3$.

Summing up k from 1 to n , simplify to get $2n - 1 < a_n^2 < 3n - 2$. Apply $n = 100$.

19. $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(x) = \begin{cases} 3x+1, & x \text{ odd,} \\ \frac{x}{2}, & x \text{ even.} \end{cases}$$

Show that for any natural number a_0 , the iterates $f^{(n)}(a_0)$ will eventually end in 1.

19. This is still yet not solved.

20. Define an infinite binary sequence as follows: start with 0 and repeatedly replace each 0 by 001 and each 1 by 0.

(a) Is the sequence periodic? (No).

(b) What is the 1000th place of the sequence?

(c) What is the place number of the 10000th one in the sequence?

(d) Find a formula for the positions of the ones (3, 6, 10, 13, ...) and a formula for the positions of the zeros. ($f(n) = \lfloor (2 + \sqrt{2})n \rfloor$ and $g(n) = \lfloor \sqrt{2}n \rfloor$.)

21. $a_0 = a_1 = 1, a_{n+1} = a_{n-1}a_n + 1, n \geq 1$. Show that $4 \nmid a_{1964}$. (Try modulo 4).

22. $a_1 = a_2 = 1, a_n = (a_{n-1}^2 + 2) / a_{n-2}, n \geq 3$. Show that all a_i are integers. (Try to show $\frac{a_{n+1} + a_{n-1}}{a_n}$ are constant, hence $a_{n+1} = 4a_n - a_{n-1}$.)

23. $a_1 = \sqrt{2}, a_{n+1} = \sqrt{2}^{a_n}$. Find its limit. (Increasing and bounded above by 2, limit is 2.)

(No. 20 to 23 drawn from A. Engel, Problem Solving and Strategies, Springer 1998).

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